



Advanced Agentic AI – Production Engineering

For Walmart Global Tech · Global Tech Academy

Date: 16 June 2026 | Version: India – 20 Hours (Advanced Only)

Program at a Glance

Walmart engineers already know the tools. The next step is production-grade judgment – selecting the right approach, shipping with confidence, and defending the decision when it matters.

20h	5 Modules	~65% Hands-on
Total duration (5 × 4 hrs)	Single advanced track	Lab over lecture

This is a single-phase, advanced-only programme. It does not repeat foundational topics. Every session assumes participants already know the basics of LLMs, prompt engineering, and RAG – and moves straight into architectural reasoning, trade-off analysis, and operational readiness.

Audience

Built for **AI Engineers and Applied-AI Practitioners** – software engineers and data scientists who have shipped GenAI prototypes and are now ready to build production systems that must survive review, scale, and cost governance.

Dimension	Profile
Experience	4–10 years; past prototyping, now moving to production
Current proficiency	Strong Python; comfortable with REST APIs and foundation models at consumer level; has built a RAG or agent demo
Environment	Mixed cloud (GCP / Azure); code-review gates; shipping-cadence pressure
Identity	Builders at scale – pragmatic, shipping-focused, eager to make defensible choices

Prerequisite Knowledge

To keep the programme advanced and engaging, the following topics are **excluded**. Participants must complete them as pre-work or via recorded sessions before joining.

- Prompt Engineering Fundamentals
- Introduction to LLMs & Transformer Mechanics
- Basic GenAI Concepts (embeddings, temperature, context window)
- Introduction to Agents & Vector Databases
- Basic LangChain / LangGraph Tutorials
- Basic Python Coding Exercises

If a participant cannot differentiate between a prompt, a workflow, and a fine-tune, they should not attend this programme.

Tools Required – Taught as Selection Judgement

Participants will use the following tools, but the focus is **not on how to use them** – it is on **when and why to choose one over another**.

Layer	Tool(s)
Language & runtime	Python (Java, .NET, JS referenced for comparison)
Orchestration frameworks	LangChain, LangGraph – taught comparatively, including when raw Python beats a framework
Agent interoperability	MCP (Model Context Protocol), function-calling / REST APIs – compared as core interoperability options

Embeddings & vector store	pgvector / Chroma / FAISS (lab); managed Pinecone for comparison
Evaluation & observability	Langfuse / Phoenix – golden-set regression, drift, cost / latency
Fine-tuning	LoRA / QLoRA via PEFT + Hugging Face; GPU-enabled Cloud Lab

Programme Structure – 5 Modules × 4 Hours

Each module is a 4-hour block with ~65% **hands-on lab** and ~35% interactive lecture / architecture discussion. The progression is deliberate: from decision-making → orchestration → production readiness → evaluation & debugging → economics & defence.

Module 1 – Engineering Decisions for AI Systems (4 Hours)

Outcome

Participants can justify technology and architecture choices for a given business use case, and document the rationale in a trade-off note.

Topics (What & Why)

- **AI application architecture patterns** – when is an agent better than a workflow, and when is traditional software still the right answer?
- **RAG vs Fine-tuning vs Prompting** – quantified decision framework (cost, latency, quality, maintainability)
- **API vs MCP** – interoperability trade-offs; when to adopt MCP and when a REST API is sufficient
- **LangChain vs LangGraph vs Python-only** – complexity, maintainability, platform support, total cost of ownership
- **Build vs Buy** – harness systems, off-the-shelf agents vs in-house orchestration
- How to write a decision matrix that survives an architecture review

Lab (Hands-on)

- Analyse 3 real-world business use cases (provided by Walmart) and select the appropriate architecture for each.
- For each case, produce a one-page decision matrix with clear trade-off rationale.

Deliverable

Decision Matrix – a structured document that maps business requirements to technology choices, with explicit pros / cons and a defended recommendation.

Module 2 – Agent Architecture & Orchestration (4 Hours)

Outcome

Participants can design scalable agent systems using multiple orchestration patterns and compare their strengths and weaknesses.

Topics (What & Why)

- **Agent design principles** – single-agent vs multi-agent systems
- **Orchestration patterns (deep dive)**: Sequential, Router, Planner-Executor, Supervisor / Hierarchical
- **Tool integration strategies** – when to use built-in tools vs custom functions vs MCP servers
- **Agent-to-agent communication protocols** – how they connect, message passing, state sharing
- **Failure scenarios** – what happens when an agent times out, hallucinates, or loses context; retry and fallback strategies

Lab (Hands-on)

- Take one use case (e.g., customer-support automation) and implement three different orchestration approaches (sequential, router, supervisor) using the same underlying tools.
- Run the same test queries against all three and observe differences in latency, accuracy, and cost.

Deliverable

Architecture Comparison Report – a side-by-side comparison of the three approaches, with recommendations for which pattern fits which scenario.

Module 3 – Production-Ready AI Systems (4 Hours)

Outcome

Participants can move a prototype to production by applying reliability, security, and governance controls.

Topics (What & Why)

- **Production readiness checklist for GenAI** – what must be true before a release
- **Reliability engineering** – guardrails, safety layers, content moderation, hallucination detection
- **Security** – prompt-injection defence, jailbreak resistance, PII handling, exfiltration controls
- **Data governance** – data lineage, versioning, audit trails
- **Scaling & latency management** – batching, streaming, caching, model routing
- **SLA / SLO considerations** – defining and measuring success in production
- **Resilience patterns** – circuit breakers, retries, fallback models, graceful degradation

Lab (Hands-on)

- Review a flawed AI application (provided as a case study) with known security, reliability, and governance gaps.
- Conduct a structured production-readiness review, identifying critical issues and proposing fixes.

Deliverable

Deployment Readiness Assessment – a formal report that scores the application against a predefined rubric and lists mandatory fixes before launch.

Module 4 – Evaluation, Observability & Debugging (4 Hours)

Outcome

Participants can measure system quality, design evaluation harnesses, and debug agentic workflows using observability data.

Topics (What & Why) – as emphasised by Naman

- **Why evaluation matters** – moving from ad-hoc testing to rigorous, repeatable measurement
- **Evaluation frameworks** – output-level (hallucination, relevance, toxicity), retrieval-level (hit rate, MRR), and agent-level (task success, tool-call accuracy)
- **Benchmark design** – creating golden datasets, regression testing, LLM-as-judge approaches
- **Logging architecture** – what to log, how to structure logs for agentic systems (traces, spans, tool calls)
- **Tracing and observability** – using Langfuse / Phoenix to visualise agent decision paths, identify bottlenecks, and detect drift
- **Debugging agentic workflows** – step-by-step root-cause analysis when a multi-step agent fails

Lab (Hands-on)

- Build an evaluation pipeline for a RAG + agent system: run a golden test set, compute metrics, and generate a scorecard.
- Instrument the same system with tracing and produce an observability dashboard that highlights latency, cost, and error rates per step.

Deliverable

Evaluation Scorecard + Observability Dashboard Design – a documented test harness with pass / fail thresholds, plus a mock-up of the monitoring dashboard used in production.

Module 5 – AI Economics, Optimization & Architecture Review (4 Hours)

Outcome

Participants can optimise token usage and operational costs, and defend their entire architecture in a simulated review board.

Topics (What & Why) – as emphasised by Naman

- **Token economics** – understanding cost drivers (input vs output tokens, model choice, context length)
- **Cost optimisation strategies** – context-window management, prompt compression, model routing (small vs large), caching, batch vs streaming
- **FinOps for GenAI** – setting cost budgets, governance thresholds, auto-scaling policies
- **Governance considerations** – cost allocation per team / project, chargeback models
- **Architecture review techniques** – how to prepare for a formal review, what questions to expect, how to document trade-offs
- Writing a trade-off note that another engineer can actually use six months later

Capstone (Integrated Lab)

- Teams of 2–3 design a production AI solution from scratch for a given Walmart business problem.

Deliverable

Architecture Review Board (ARB) Presentation – a formal deck that defends every major architectural decision under cost, latency, and governance pressure. Peers and instructors act as the review board, asking tough questions.

Each capstone team must present:

- **Architecture choice** – agent / workflow / traditional
- **Agent strategy** – orchestration pattern, tool selection
- **Evaluation strategy** – metrics, golden set, go / no-go criteria
- **Cost model** – projected token usage, monthly spend, optimisation levers
- **Deployment model** – CI/CD, rollback, incident response
- **Risk mitigation plan** – security, hallucinations, drift

Final Programme Outcomes

By the end of the 20 hours, every participant will be able to:

#	Outcome
1	Select the right AI architecture for a business problem and justify it with a trade-off matrix.
2	Design agentic systems using appropriate orchestration patterns (sequential, router, supervisor, etc.).
3	Build production-ready AI applications with security, reliability, and governance controls.
4	Evaluate, monitor, and debug AI systems using rigorous metrics, observability, and tracing.
5	Optimise token usage and operational costs via caching, model routing, and smart context management.
6	Defend technical decisions during formal architecture reviews, using data and documented trade-offs.

Assessment & Certification

Completion is based on active participation in labs and submission of **all 5 deliverables**.

Certification requires a passing grade on the final ARB presentation (Module 5 capstone), assessed against a rubric covering:

- Architectural reasoning
- Trade-off documentation
- Cost / latency / governance awareness
- Risk identification and mitigation

No attendance-only certificates are issued – participants must demonstrate they can ship and defend.

Schedule Overview (Suggested)

Day	Module	Duration
Day 1	Module 1 – Engineering Decisions	4 hours
Day 2	Module 2 – Agent Architecture & Orchestration	4 hours
Day 3	Module 3 – Production-Ready AI Systems	4 hours
Day 4	Module 4 – Evaluation, Observability & Debugging	4 hours
Day 5	Module 5 – AI Economics, Optimization & ARB Capstone	4 hours

Each day includes two 15-minute breaks and a 30-minute lunch break (if full-day) – adjust per local scheduling.